

Disk-onnected: Transformers Struggle to Use Their World Models

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Abstract

Large reasoning models collapse on Tower of Hanoi planning once the puzzle grows past a handful of disks, a failure documented behaviourally but never explained mechanistically. We ask where in the computation the failure lives: does the model never form a representation of the puzzle state, lose it across a long reasoning trace, or hold it intact and fail to act on it? We answer with two complementary studies on the flat-to-flat Tower of Hanoi. First, a small Transformer trained from scratch on solution traces is shown, through linear distance-matching probes and activation patching, to develop a linearly decodable, geometrically faithful representation of the disk configuration that reproduces the Sierpiński geometry of the state graph and is causally read when the model emits a solution. Second, the same probes are applied to two frontier reasoning models, Qwen3.6-27B and DeepSeek-R1-Distill-Qwen-32B, that solve the task through extended chain-of-thought. Both encode the configuration near-perfectly at the end of the prompt, yet solve only about half of instances optimally; tracking the probe through generation reveals that the representation degrades as the model reasons, decaying from a unified planning geometry into a fragmented one at the point of commitment. Restoring the clean prompt-time representation, by injecting it at inference or by training it back into the weights with a state-encoding auxiliary loss, increases optimal-solve accuracy for Qwen3.6-27B, though the same intervention fails to transfer to DeepSeek-R1-Distill. The results reframe the reported collapse: these models build a world model and then lose it. The failure is one of maintenance, not absence, and it is at least partially recoverable.

1 Introduction

Current models capable of drafting proofs, writing working code, and scoring at the frontier of graduate examinations cannot reliably solve a puzzle designed for children. Solving the Tower of Hanoi requires large reasoning models (LRMs) to move a stack of disks between three pegs, yet they are incapable of doing so: their accuracy falls off a cliff once the puzzle grows past a handful of disks, and, stranger still, their reasoning traces get shorter at exactly the point where the problem gets harder (Shojaee et al., 2025). The behaviour has been measured carefully and named the “illusion of thinking”, but this phenomenon has not been explained mechanistically. We know that these models fail. But, we do not know what inside the model is failing.

This gap motivates a debate about what these systems are and wherein their capabilities truly lie. When a sequence model trained only to predict the next token solves a planning problem, is it manipulating an internal model of the world according to the problem it is solving, or is it recombining surface statistics with no underlying state to speak of (Bender et al., 2021)? The evidence so far is split. Small Transformers trained on Othello and chess move histories do build decodable causally active board representations (Li et al., 2022; Toshniwal et al., 2022); maze-solving networks encode walls and paths (Spies et al., 2025). But these demonstrations so far study a model that emits one move per forward pass. None study a reasoning model that discusses, justifies and contradicts its way to an answer through thousands of tokens of chain-of-thought. This is precisely the setting in which such

failures occur. The two literatures, the one documenting collapse and the other documenting world models, have never been combined into a single coherent study.

Throughout this work, we consider planning failures to be a genuine mechanistic claim, asking where they lie. For a model that collapses on a planning problem, there are three competing explanations that call for opposite solutions. (1) The model may never form a representation of the puzzle state, in which case the apparent reasoning is a facade. (2) It may form the representation but lose it somewhere across a long reasoning trace, in which case the failure is one of maintenance and the fix is a richer training signal over states. (3) Or it may hold the state intact and simply fail to act on it, in which case the failure is one of decoding and the fix lies in how the model generates. Behavioural benchmarks cannot tell these apart. Only analyzing model internals can.

We pry them apart with two different designs. First, a small Transformer trained from scratch on Tower of Hanoi solutions gives a fully controlled setting in which the existence, geometry, format, and causal role of any internal state representation can be established with probing, sparse autoencoders, and activation patching. Second, two frontier reasoning models, Qwen3.6-27B and DeepSeek-R1-Distill-Qwen-32B, are put through the same lens to test whether those findings survive at scale and under chain-of-thought. We can watch the representation evolve token by token as the model generates.

The answer that emerges is sharper than whether the model can or cannot reason. A geometrically structured world model does emerge in models separated by orders of magnitude in size. We find that it is initially decodable, then degrades as the model reasons, from a unified planning geometry into a fragmented one at the moment the model commits to its moves. Furthermore, in one frontier model, deliberately restoring that representation, either by injecting it at inference or by training it back into the weights, recovers the lost accuracy. The illusion of thinking, it turns out, is not that the model never thinks; it is that the model forgets what it knew.

We test three hypotheses, the first two on a toy Transformer trained from scratch and the third on frontier reasoning models. **H1:** a simple autoregressive Transformer trained on flat-to-flat Tower of Hanoi traces develops a linearly decodable representation of the disk configuration. **H2:** this representation is causally implicated in the model’s solution. **H3:** in large reasoning models the representation degrades over autoregressive generation, and this degradation, rather than its absence, is what drives failure.

2 Related work

Planning and its failure in reasoning models. Reasoning models exhibit a counterintuitive failure mode on long-horizon planning problems. Their accuracy collapses to zero once compositional depth exceeds a model-specific threshold, while their reasoning traces grow shorter rather than longer at the point of failure (Shojaee et al., 2025). Behavioural evidence of this collapse is now well documented, but its mechanistic origin remains unclear. A separate line of work has shown that Transformer sequence models can develop internal representations of the latent state they predict over, even when trained only on next-token prediction (Li et al., 2022; Toshniwal et al., 2022). World-model emergence has been demonstrated almost exclusively in small Transformers trained on games (Li et al., 2022; Toshniwal et al., 2022) or path prediction in mazes (Spies et al., 2025). Whether such world models extend to large reasoning models producing extended reasoning traces is still yet to be discovered. There is a paucity in research on the question of whether a reasoning model’s internal representation of a planning problem is what determines its success.

Probing, decomposing, and intervening on representations. We test for encoded world models with linear probes (Belinkov, 2022), motivated by the linear representation hypothesis (Mikolov et al., 2013; Park et al., 2023) and remain mindful that some features are nonlinear or genuinely multi-dimensional (Csordás et al., 2024; Engels et al., 2025); the unified-versus-factored format we observe is quantified through principal angles between probe subspaces (Björck & Golub, 1973). Decodability does not confirm that models use this representation, therefore we establish causal involvement by activation patching (Vig

et al., 2020; Meng et al., 2022; Wang et al., 2022) and by activation steering (Turner et al., 2023; Subramani et al., 2022; Hernandez et al., 2023). The closest methodological precedent, sequence-only training combined with probing and causal edits (Li et al., 2022; Spies et al., 2025), is exactly the combination we carry from a toy Transformer to frontier reasoning models, and then turn from diagnosis into a state-encoding training signal that recovers accuracy.

3 Method

The investigation has two streams. The first studies a toy autoregressive Transformer trained from scratch on flat-to-flat Tower of Hanoi traces, where the existence, geometry, and causal role of any internal state representation can be characterised in detail. The second applies the same probing techniques to frontier large reasoning models (LRMs) that solve the task through extended chain-of-thought. We adopt the *flat-to-flat* formulation, in which the initial and goal states are arbitrary valid configurations rather than full towers. The optimal solution depends on the pair (s_I, s_G) and cannot be produced by a memorised recursive template. Optimal solutions are found by breadth-first search on the Hanoi state graph, motivating the models to form an internal understanding of the state space to optimally solve the problem

3.1 Toy transformer: probes and causal intervention

The toy model is a GPT-2-style decoder-only Transformer (Radford et al., 2019) trained on tokenised flat-to-flat traces with a cross-entropy loss masked to the move tokens. The state of all four disks is read out at the separator token (SEP) that sits between the problem and the solution.

Distance-matching probe. To test whether the joint configuration is encoded, we train a linear probe $f_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^2$ that maps a hidden state to a 2D embedding whose pairwise Euclidean distances match the graph distances d_G between configurations:

$$\mathcal{L}_{\text{probe}}(\phi) = \frac{1}{|\mathcal{S}|^2} \sum_{s, s' \in \mathcal{S}} (\|f_\phi(h_s) - f_\phi(h_{s'})\|_2 - d_G(s, s'))^2, \quad (1)$$

where $\mathcal{S}$ is the set of all 81 four-disk configurations. Low loss means the joint state is recoverable through a single linear map.

Per-disk probes and subspace analysis. To characterise how the state is organised, we train one logistic probe per disk k (weight matrix $W_k \in \mathbb{R}^{3 \times d}$) predicting its peg, and treat the row span of W_k as that disk’s encoding subspace. The *principal angle* between two such subspaces (the smallest angle between any pair of unit vectors drawn from them, via the SVD of their orthonormal bases) measures representational format. Angles near 0° indicate a *unified* encoding in a shared subspace, angles near 90° a *factored*, mutually orthogonal one.

Activation patching. Decodability is not use. To test causal involvement we patch (Meng et al., 2022) the SEP activation: running the model on a corrupted configuration s_{corrupt} but substituting the SEP activation from a clean pass on s_{clean} , and classifying the emitted sequence as FULL TRANSFER (matches s_{clean}), PARTIAL (some disks follow the donor), UNCHANGED (matches s_{corrupt}), NOVEL (optimal for a third configuration), or DISRUPTED (solves neither).

3.2 Large reasoning models: probing through generation, steering, fine-tuning

For the LRMs, the distance-matching and per-disk probes are applied unchanged to at three positions in a single forward pass: **Position A**, the final prompt token (the LRM analogue of SEP, before any reasoning); **Position B**, the last token before the emitted move list (the commitment point, after the full chain-of-thought); and **Position C**, the move-emission tokens (one sample per move).

Activation steering. To test causal use we steer (Turner et al., 2023). We cache the layer- ℓ Position-A activation $h_{\text{prompt}}(s)$ for each of the 81 configurations and their mean $\bar{h}$. At each move-emission position, using an externally simulated tracker to recompute the true post-move configuration s_t , we add a unit steering vector toward that configuration’s clean activation:

$$\tilde{h}^{(\ell)} = h^{(\ell)} + \alpha \cdot \frac{h_{\text{prompt}}(s_t) - \bar{h}}{\|h_{\text{prompt}}(s_t) - \bar{h}\|_2}, \quad (2)$$

with injection strength α swept. Recovery of accuracy under steering implicates mid-generation degradation as the cause of failure.

State-encoding loss. As we want to reinforce the capability of LRMs without external aids, we fold the same correction into the weights. At each supervised move-emission position t with post-move configuration s_t , the layer- ℓ training target is the steering-displaced activation

$$h_{\text{target}}^{(\ell)}(s_t) = h_{\text{baseline}}^{(\ell)} + \alpha (h_{\text{prompt}}^{(\ell)}(s_t) - \bar{h}^{(\ell)}), \quad (3)$$

where $h_{\text{baseline}}^{(\ell)}$ is the frozen pretrained activation. A LoRA adapter is trained with the language-modelling loss plus an auxiliary term that pulls the current activation toward this target over a contiguous mid-band of layers $\mathcal{B}$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LM}} + \frac{\lambda}{|\mathcal{B}| N_{\text{sup}}} \sum_{\ell \in \mathcal{B}} \sum_{t \in T_{\text{sup}}} \frac{1}{d} \|h_{\text{current}}^{(\ell)}(t) - h_{\text{target}}^{(\ell)}(s_t)\|_2^2, \quad (4)$$

with T_{sup} the supervised positions ($N_{\text{sup}} = |T_{\text{sup}}|$) and d the hidden dimension. At inference the trained adapter runs with no hook, probe, or external tracker.

3.3 Evaluation

Probe quality is reported by the Spearman (ρ) and Pearson (r) correlations between predicted and true pairwise distances, and nearest-state retrieval accuracy (the fraction of configurations whose nearest embedding neighbour is the true nearest graph neighbour). A high- ρ with lower- r signals an order-preserving but non-uniformly scaled embedding. Solutions are classified OPTIMAL, SUBOPTIMAL (legal, goal-reaching, over-long), INCORRECT (legal, goal-missing), or ILLEGAL (a rule violation or unparseable output).

4 Experimental setup

Toy Transformer. All experiments use the $n = 4$ Tower of Hanoi, whose 81 valid configurations form a Sierpiński triangle and yield 6,480 ordered (s_I, s_G) pairs with $s_I \neq s_G$, split 80/20 into 5,184 training and 1,296 validation problems. The model has 6 layers, hidden dimension 128, 4 heads and is trained for 50 epochs. It reaches 99.2% token-level and 93.2% sequence-level validation accuracy. Probes are trained at each of the 6 layers; move-token probes use the 49,450 move-token activations from optimally solved problems.

Large reasoning models. We probe two open-weight reasoning models that emit explicit <think> chains: Qwen3.6-27B and DeepSeek-R1-Distill-Qwen-32B (both 64 layers, hidden dimension 5,120). Probing is applied at 17 layer indices (layer 1 and every fourth from 4 to 64). Problem instances are the 81 reachable configurations paired with a fixed goal ($N = 81$ for Position A). For baselines we follow the protocol of Shojaaee et al. (2025), varying only the generation token budget; sampling settings are in Appendix E.

5 Toy transformer: an emergent, causal world model

5.1 The joint state is linearly decodable at SEP

A linear distance-matching probe at the SEP token recovers the joint four-disk configuration with high fidelity (Table 1): Spearman correlation reaches 0.94 at the best layer (layer 5) and

Table 1: SEP-token distance-matching probe at epoch 50, evaluated on all 81 configurations, by layer. The joint configuration is linearly decodable, peaking at layer 5.

Layer	Spearman ρ	Pearson r
5	0.938	0.902
4	0.936	0.899
6	0.934	0.898
3	0.931	0.895
2	0.921	0.878
1	0.829	0.814

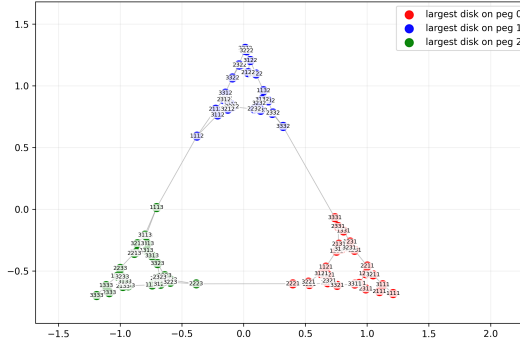


Figure 1: 2D output of the distance-matching probe on the raw residual stream at layer 5, at epoch 50. Each point is one of the 81 four-disk configurations, coloured by the largest disk’s position; grey edges connect graph-adjacent configurations. The three largest-disk sub-triangles separate into the Sierpiński geometry, with cross-cluster gaps wider than the within-cluster edges. Earlier epochs are shown in Appendix B.

183 Pearson 0.90, with the geometry assembled by middle depth. The persistent $\rho-r$ gap of 3–4
 184 points indicates an embedding that preserves the order of graph distances but inflates the
 185 separation between the three sub-triangles defined by the largest disk’s position, exactly the
 186 recursive cost structure of the puzzle. The converged probe lays the 81 configurations out in
 187 the Sierpiński triangle of the state graph (Figure 1), a geometry that sharpens over training
 188 (Appendix B). Probing individual move-token activations, by contrast, fails to recover the
 189 joint state (best $\rho = 0.78$ with a 15-token window; Appendix C), even though the same state
 190 is perfectly decodable at SEP a few positions earlier, motivating the per-disk analysis below.
 191 This supports **H1**.

192 5.2 A unified-to-factored shift between planning and execution

193 Per-disk probes show the state is present at both positions but organised differently (Table 2).
 194 At SEP every disk is recovered perfectly; at move tokens the two small disks remain
 195 near-perfect while the larger disks degrade (to 90.7% and 79.3%), as the model loses track
 196 of infrequently moved disks. Principal angles make the format explicit. Between disk
 197 subspaces, they average 54.7° at SEP (a *unified*, overlapping encoding that supports the joint
 198 Sierpiński geometry) but rise to 76.5° at move tokens (a *factored*, near-orthogonal encoding
 199 from which each disk can be read independently). The joint distance structure is thus not
 200 the sum of the per-disk parts: it lives in the overlapping geometry that the distance probe
 201 captures and the per-disk probes discard.

Table 2: Per-disk classification accuracy (best layer, layer 5). Perfect at SEP; the larger disks degrade at move tokens.

Position	Disk 0	Disk 1	Disk 2	Disk 3	Mean
SEP	100.0%	100.0%	100.0%	100.0%	100.0%
Move tokens	99.98%	99.96%	90.70%	79.25%	92.47%

Table 3: Activation patching at the SEP token across all donor–recipient validation pairs. PARTIAL transfer dominates; DISRUPTED falls to zero with depth.

Layer	Full	Partial (mean K)	Unchanged	Novel	Disrupted
4	5.75%	63.89% ($K=1.46$)	17.56%	0.37%	12.42%
5	6.00%	71.77% ($K=1.45$)	16.49%	0.11%	5.63%
6	5.89%	78.74% ($K=1.44$)	15.37%	0.00%	0.00%

202 5.3 The representation is causally used

203 Activation patching closes the loop from correlation to causation (Table 3). Substituting a
 204 donor’s SEP activation transfers state to the recipient’s output in a large majority of pairs.
 205 PARTIAL transfer dominates at every layer (64–79%), where $K \approx 1.45$ of four disks on
 206 average are copied from the donor’s solution. FULL transfer is present ($\sim 6\%$), and NOVEL
 207 outcomes are rare ($\leq 0.4\%$), so outputs stay tied to donor or recipient. The DISRUPTED rate
 208 falls with depth, from 12.4% at layer 4 to zero at layer 6, i.e. late-layer SEP activations admit
 209 clean replacement. The influence is therefore causal but graded, supporting H2.

210 6 Large reasoning models: emergence, degradation, restoration

211 6.1 Flat-to-flat is hard, and depends on reasoning length

212 While the standard tower-to-tower task is saturated for current reasoning models (Ap-
 213 pendix F), the flat-to-flat formulation is far harder (Table 4). Even the strongest models
 214 solve only about half of instances optimally, and accuracy falls steeply with disk count. The
 215 final two rows re-evaluate the open-weight models under the much tighter output budget
 216 later used for fine-tuning. This isolates how much accuracy depends on reasoning length.
 217 Restricting Qwen3.6-27B to that budget collapses its optimal-solve rate from 51% to 14%,
 218 despite an unchanged Position-A representation: the model that solves half the instances
 219 with a mean of $\approx 19,000$ generated tokens solves a seventh of them when capped. No-
 220 tably Qwen3.6-27B generates longer traces than DeepSeek-R1-Distill-Qwen-32B on average
 221 ($\approx 19,000$ vs $\approx 14,000$ tokens), a fact we return to for the steering asymmetry.

222 6.2 Position A: a world model at the end of the prompt

223 At Position A the distance-matching probe recovers the configuration to essentially perfect
 224 accuracy in both models (full per-layer results in Table 8): from middle depth onward
 225 Spearman reaches 0.935 and nearest-state accuracy 1.00, identical to the toy SEP token,
 226 and the same $\rho-r$ asymmetry is preserved. The representation is largely absent at the first
 227 layer, but is assembled by layer 8, then plateaus; DeepSeek-R1-Distill-Qwen-32B reaches
 228 $\rho = 0.936$ with perfect retrieval by middle depth as well. The fact that a 27B pre-trained
 229 reasoning model and a 6-layer Transformer trained from scratch encode the configuration
 230 with identical fidelity is the strongest evidence that the world model is a property of the
 231 task, not of scale or training, extending H1 to frontier scale.

Table 4: Baseline flat-to-flat accuracy (instances solved optimally) on 100 sampled instances per model. **Max tokens** is the output budget (32k caps the answer only, with provider-side reasoning beyond it; the last two rows use a hard cap: 4096 for $n = \{3, 4\}$ and 8192 for $n = 5$).

Model	Max tok	3 disks	4 disks	5 disks	Total
DeepSeek-R1	32k	24/34	12/33	4/33	40/100
Kimi-K2-Think	32k	19/34	3/33	3/33	25/100
gpt-oss-120b	32k	26/34	12/33	13/33	51/100
DeepSeek-R1-Distill-Qwen-32B	32k	6/34	0/33	1/33	7/100
Qwen3.6-27B	32k	28/34	14/33	9/33	51/100
DeepSeek-R1-Distill-Qwen-32B	4–8k	14/34	1/33	0/33	15/100
Qwen3.6-27B	4–8k	8/34	2/33	4/33	14/100

Table 5: Probing Qwen3.6-27B through generation. D0–D3 are per-disk probe accuracies.

Position	Layer	Spearman	Acc	D0	D1	D2	D3
A	24	0.935	1.00	0.75	0.75	0.80	0.90
A	36	0.935	1.00	0.59	0.69	0.75	0.95
A	48	0.935	1.00	0.48	0.57	0.70	0.95
B	24	0.915	0.83	0.33	0.37	0.42	0.43
B	36	0.915	0.75	0.33	0.33	0.41	0.34
B	48	0.920	0.82	0.34	0.36	0.42	0.43
C	24	0.802	0.41	0.80	0.70	0.75	0.89
C	36	0.813	0.64	0.58	0.62	0.60	0.84
C	48	0.847	0.69	0.49	0.58	0.55	0.78

6.3 Positions B and C: degradation through generation

A faithful prompt-time representation does not translate into a solution. With a maximal reasoning budget, Qwen3.6-27B solves only 51% optimally and the distilled model far fewer. Tracking the probe through generation locates the gap (Table 5). At the commitment point (Position B) the joint configuration is not as easily decodable. For Qwen3.6-27B the global geometry is roughly preserved ($\rho \approx 0.92$) but nearest-state accuracy drops and the per-disk classifiers collapse to near chance. DeepSeek-R1-Distill-Qwen-32B degrades further on both axes (ρ to 0.74–0.81). During move emission (Position C) the trend of a more factored, per-disk encoding partially returns. This is the large-model counterpart of the toy unified-to-factored shift: a *unified geometry* where planning happens, a *factored* one where execution happens. The degradation appears even on solved problems, therefore maintaining a clean configuration across a long trace is hard even when the model ultimately succeeds, which supports **H3**.

6.4 Restoring the world model recovers performance

If world model degradation is the bottleneck, restoring the clean Position-A representation at move-emission positions should help. For Qwen3.6-27B it does (Table 6). Steering is applied only to the problems the model fails unaided. Even the weakest setting converts 44% of failures into optimal solutions. At the best strength ($\alpha = 5$), it lifts the optimal-solve count on the 81 configurations from 33 (41%) to 63 (78%). The effect is non-monotonic in α , i.e. too large a value destabilises generation. The residual failures are rule-violating moves, not parse errors: well-formed but wrong. This is direct causal evidence that the degraded world model limits the model, supporting **H3**.

For DeepSeek-R1-Distill-Qwen-32B the same intervention does very little: the outcome distribution is identical across every α , and even an extreme value ($\alpha = 500$, verified large

Table 6: Activation steering at layer 36, on the problems each model fails unaided (Qwen: $N=48$; DeepSeek: $N=70$).

Model	α	Optimal	Suboptimal	Incorrect	Illegal
Qwen3.6-27B	0.5	21	10	1	16
	1	22	6	3	17
	2	18	13	1	16
	5	30	5	3	10
	10	21	5	5	17
DeepSeek-R1-Distill-Qwen-32B	0.5–10	2	2	1	65

256 enough to flip the output logit) yields only parse errors and illegal moves. The natural
 257 explanation is output format: DeepSeek-R1-Distill-Qwen-32B frequently fails to emit a
 258 parseable move list at all (35 of 70 outputs), so a large share of its failures occur at a stage a
 259 state-restoring nudge cannot reach.

260 6.5 Restoration in the weights: a state-encoding loss

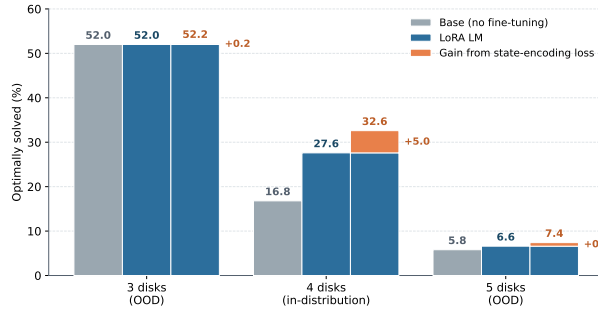


Figure 2: Optimally solved problems for Qwen3.6-27B: the base model with no fine-tuning (grey), and after one epoch of LoRA fine-tuning under the language-modelling baseline vs. the added state-encoding loss (layer 36, $\alpha = 0.2$). The coral segment is the gain from the state-encoding loss. $n = 4$ is in-distribution; $n = 3, 5$ are out-of-distribution.

261 We apply the same correction into the weights using a LoRA adapter as an auxiliary state-
 262 encoding loss (Section 3.2); it improves the optimal-solve count at every disk count over
 263 a language-modelling-only baseline, on 500 held-out problems per disk count (Figure 2):
 264 the largest gain is in-distribution ($n = 4$: 138 $\rightarrow$ 163) at no cost out-of-distribution ($n = 3$:
 265 260 $\rightarrow$ 261; $n = 5$: 33 $\rightarrow$ 37). Where steering is an inference-time control, here the same
 266 direction serves as a training signal that strengthens the state encoding into the weights,
 267 removing the need for any external state-tracker at deployment.

268 7 Discussion

269 The two streams lead to one narrative: a geometrically structured world model emerges by
 270 the end of the prompt in models separated by orders of magnitude in size, degrades over
 271 the course of generation, and recovers solution accuracy when restored, at inference or in
 272 the weights.

273 **An emergent, causal world model that is not an artifact of scale.** The distance-matching
 274 probe recovers the joint configuration, and reproduces the Sierpiński geometry of the state
 275 graph, at the toy SEP token and at the final prompt token of both frontier models with near-
 276 identical fidelity, down to the shared Spearman–Pearson gap that signals inflated separation

between the largest-disk sub-triangles (Tables 1 and 8). That a 6-layer Transformer trained from scratch and a 27B reasoning model encode the state alike argues the world model is a property of the task, not of scale or training regime. We extend the emergent-world-model line (Li et al., 2022; Toshniwal et al., 2022; Spies et al., 2025) from one-move-per-pass game settings into planning solved through extended chain-of-thought. Activation patching closes the loop from correlation to causation. Substituting a donor’s SEP activation transfers state to the recipient’s output in a large majority of pairs (Table 3), so the representation is read, not merely present.

Degradation through generation reframes the collapse. A faithful prompt-time representation does not yield a solution: both models recover the state near-perfectly at Position A yet solve only about half of flat-to-flat instances optimally (Tables 4 and 5). Tracking the probe through generation locates the gap, the joint configuration becomes hard to read at the commitment point and a more factored, per-disk encoding partially returns during emission, the large-model counterpart of the toy unified-to-factored shift. This sharpens Shojaei et al. (2025): the collapse is, at least in part, a failure to maintain a representation the model demonstrably had, not the absence of one. Restoring the clean Position-A activation at move-emission positions nearly doubles Qwen3.6-27B’s optimal-solve count (Table 6), direct causal evidence that the degraded world model is the bottleneck. As in Othello-GPT (Li et al., 2022), the steered model acts on the injected configuration, emitting well-formed move lists with no format errors at any α ; its residual failures are illegal *moves*, not malformed output. Folding the same direction into a LoRA adapter as an auxiliary state-encoding loss improves accuracy at every disk count, largest in-distribution and at no out-of-distribution cost (Figure 2).

Steering does not transfer The intervention does essentially nothing for DeepSeek-R1-Distill-Qwen-32B, despite its near-identical Position-A probe (Table 6). The cleaner differentiator between the two models is output format, DeepSeek frequently emits no parseable move list at all, so many failures occur where a state-restoring nudge cannot reach. A representational mismatch, the load-bearing layer-36 direction for Qwen not being DeepSeek’s own state code at emission, remains possible; separating the two is the most important open question we leave to future work.

Limitations. Experiments use a single puzzle at a single size ($n=4$), with only small out-of-distribution checks at $n=3, 5$; Tower of Hanoi’s clean recursive structure is what makes its graph a convenient probe target. Whether the story of emergence then degradation, and finally restoration, holds for less geometric state spaces is untested. The two frontier models are both Qwen-derived, so transfer should be read within that caveat. A supervised probe can fit structure the model does not use, we mitigate this with patching and steering, which test causal use directly. Several figures rest on single seeds, and the state-encoding loss was evaluated at one hyperparameter setting (layer band, α , auxiliary weight), so its gains are a lower bound. The steering method also relies on an external state-tracker that recomputes the true state after each move, feasible here but not obviously scalable to tasks where ground-truth state is expensive or ambiguous. The fine-tuning and steering runs also used a constrained output-token budget for tractability; since flat-to-flat accuracy depends on reasoning length (Table 4), the reported solve rates and the steering and fine-tuning gains are likely conservative. Finally, the technique that restores a correct world model could steer one toward a non-benign target, a capability whose governance deserves attention as such methods mature.

8 Conclusion

We asked where a reasoning model’s planning failure lives, and found it is neither an absent world model nor a pure decoding fault but a failure to maintain a representation the model demonstrably forms. A linear distance-matching probe recovers the Tower of Hanoi configuration, in faithful Sierpiński geometry. We find this structure in a toy Transformer’s SEP token and in the final prompt token of two frontier reasoning models alike (H1). Activation patching shows the toy representation is causally read rather than

merely present (H2). Furthermore, tracking the probe through generation shows the frontier representation is near-perfect at the end of the prompt, degrades at the commitment point, and returns only in factored form during emission (H3). Restoring it, either by steering at inference or through an auxiliary state-encoding loss in the weights, recovers much of the lost accuracy for Qwen3.6-27B, though not for DeepSeek-R1-Distill-Qwen-32B, whose unparseable outputs leave the intervention nothing to correct. The reported collapse is thus better read as the model forgetting what it knew than as an inability to represent the problem, a reframing that points mitigation toward maintaining state across a long trace rather than toward more inference-time search.

References

- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.
- Emily M Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*, pp. 610–623, 2021.
- Åke Björck and Gene H Golub. Numerical methods for computing angles between linear subspaces. *Mathematics of computation*, 27(123):579–594, 1973.
- Róbert Csordás, Christopher Potts, Christopher D Manning, and Atticus Geiger. Recurrent neural networks learn to store and generate sequences using non-linear representations. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pp. 248–262, 2024.
- Joshua Engels, Isaac Liao, Eric J Michaud, Wes Gurnee, and Max Tegmark. Not all language model features are linear. 2025.
- Evan Hernandez, Belinda Z Li, and Jacob Andreas. Inspecting and editing knowledge representations in language models. *arXiv preprint arXiv:2304.00740*, 2023.
- Kenneth Li, Aspen K Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Emergent world representations: Exploring a sequence model trained on a synthetic task. *arXiv preprint arXiv:2210.13382*, 2022.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. *Advances in neural information processing systems*, 35:17359–17372, 2022.
- Tomáš Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 conference of the north american chapter of the association for computational linguistics: Human language technologies*, pp. 746–751, 2013.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*, 2023.
- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- Parshin Shojaee, Iman Mirzadeh, Keivan Alizadeh, Maxwell Horton, Samy Bengio, and Mehrdad Farajtabar. The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity. *arXiv preprint arXiv:2506.06941*, 2025.
- Alexander F Spies, William Edwards, Michael Ivanitskiy, Adrians Skapars, Tilman Räuher, Katsumi Inoue, Alessandra Russo, and Murray Shanahan. Transformers use causal world models in maze-solving tasks. In *ICLR 2025 Workshop on World Models: Understanding, Modelling and Scaling*, 2025.

- 376 Nishant Subramani, Nivedita Suresh, and Matthew E Peters. Extracting latent steering
 377 vectors from pretrained language models. In *Findings of the Association for Computational*
 378 *Linguistics: ACL 2022*, pp. 566–581, 2022.
- 379 Shubham Toshniwal, Sam Wiseman, Karen Livescu, and Kevin Gimpel. Chess as a testbed
 380 for language model state tracking. In *Proceedings of the AAAI Conference on Artificial*
 381 *Intelligence*, volume 36, pp. 11385–11393, 2022.
- 382 Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J Vazquez, Ulisse
 383 Mini, and Monte MacDiarmid. Steering language models with activation engineering.
 384 *arXiv preprint arXiv:2308.10248*, 2023.
- 385 Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer,
 386 and Stuart Shieber. Investigating gender bias in language models using causal mediation
 387 analysis. *Advances in neural information processing systems*, 33:12388–12401, 2020.
- 388 Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt.
 389 Interpretability in the wild: a circuit for indirect object identification in gpt-2 small. *arXiv*
 390 *preprint arXiv:2211.00593*, 2022.

391 A Unsupervised dimensionality reduction

392 As a probe-free check, SEP-token activations for all 81 start states (fixed goal) were projected
 393 to 2D with PCA, t-SNE, and UMAP at every layer (Figure 3). None of the eighteen panels
 394 reproduces the Sierpiński geometry; at best a coarse grouping by the largest disk’s position
 395 emerges. The model does not store the state space in a representation whose *raw* projected
 396 geometry is Sierpiński-like, motivating the supervised distance-matching probe.

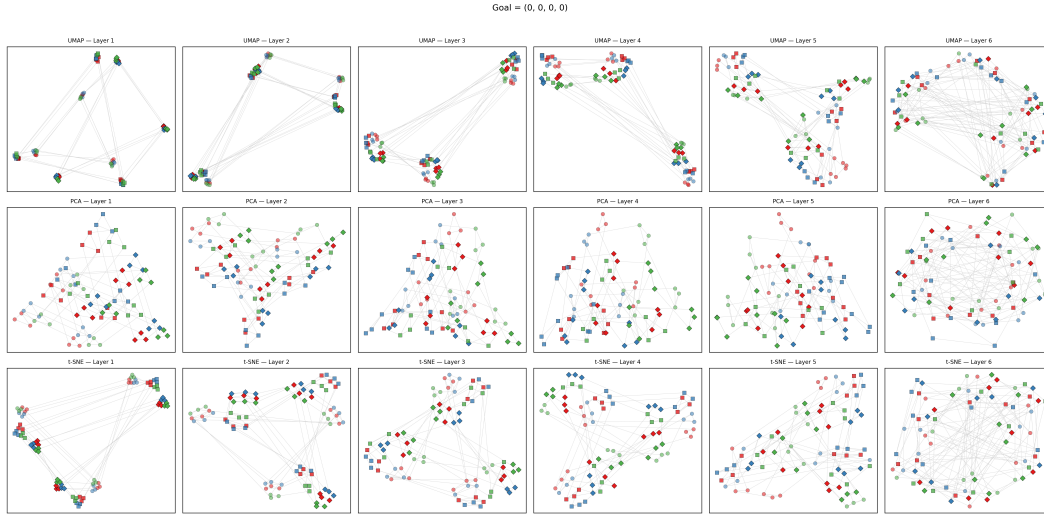


Figure 3: Two-dimensional projections of SEP-token hidden states across the three techniques (rows) and six layers (columns), coloured by largest-disk position. No panel recovers the Sierpiński structure.

397 B Probe geometry over training

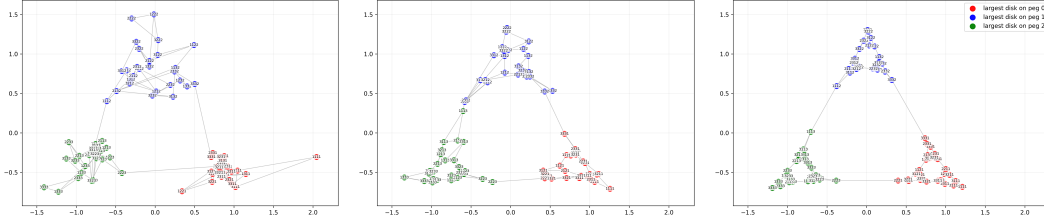


Figure 4: 2D output of the distance-matching probe at layer 5 over training (epochs 5, 25, 50). The Sierpiński structure sharpens as the three largest-disk sub-triangles separate.

398 C Move-token probes

399 The joint state is not recoverable from individual move-token activations. Across four
 400 probe variants the best result is a 15-token multi-position window at $\rho = 0.781$ and 7.6%
 401 nearest-state accuracy, far below the SEP token’s $\rho = 0.938$ at 100% retrieval (Table 7).

Table 7: Best configuration of each move-token probe (selected by Spearman), with the SEP linear probe for reference.

Probe	Layer	Nearest-state acc.	Spearman ρ
Linear (pooled)	4	1.5%	0.295
Per-step controlled	3	1.0%	0.610
Nonlinear (MLP)	4	4.1%	0.668
Multi-position ($K=15$)	3	7.6%	0.781
SEP token (linear, ref.)	5	100%	0.938

402 D Sparse autoencoder analysis

403 To test whether the move-token state is in superposition, sparse autoencoders were
 404 trained on residual-stream activations at layers 4–5 (expansion $\{2, 4, 8, 16\}$, L^1 penalty
 405 $\lambda \in \{0.5, 1, 2, 4\}$) and the distance probe re-trained on the latents. At SEP the SAE latents
 406 and raw activations give nearly identical probe quality ($\rho = 0.935$ either way); at move
 407 tokens the SAE roughly doubles nearest-state accuracy but absolute values stay very low.
 408 The SAE offers no evidence of substantial superposition: the structure the linear probes
 409 recover from the raw residual stream is the same structure the SAE recovers (Figure 5).

410 E Generation settings

411 Reasoning traces are generated with nucleus sampling ($T=1.0$, top- p 0.95, top- k 20).

412 F Tower-to-tower baselines

413 On the standard tower-to-tower task all frontier reasoning models solve near-perfectly across
 414 $n \in \{3, 4, 5\}$ (DeepSeek-R1 24/25, Kimi-K2-Think 23/25, gpt-oss-120b 25/25, Qwen3.6-27B
 415 24/25), confirming the task is saturated; the distilled DeepSeek-R1-Distill-Qwen-32B is the
 416 exception at 1/25.

417 G Full Position A probing

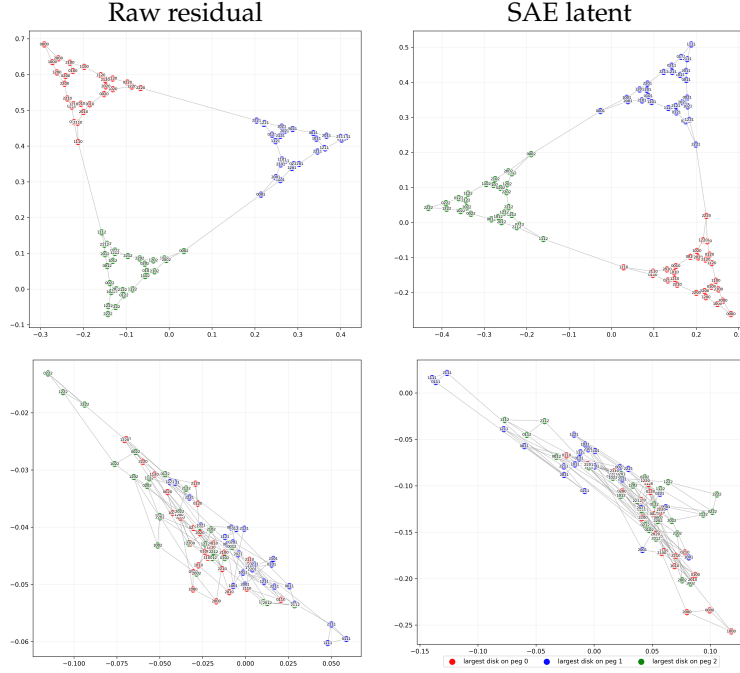


Figure 5: Distance-matching probe layouts at layer 5: raw residual stream (left) vs. SAE latents (right); SEP (top, raw $\rho=0.935$ /SAE $\rho=0.935$) and move tokens (bottom, raw $\rho=0.136$ /SAE $\rho=0.140$).

Table 8: Position A (final prompt token) probing, three best and three worst layers by Spearman. Both models recover the configuration near-perfectly from middle depth. DS-32B denotes DeepSeek-R1-Distill-Qwen-32B.

Model	Layer	Spearman	Pearson	NS acc.
Qwen3.6-27B	36	0.935	0.903	1.000
	32	0.935	0.903	1.000
	28	0.935	0.903	1.000
	8	0.897	0.887	1.000
	4	0.740	0.703	0.383
	1	0.714	0.666	0.235
DS-32B	8	0.936	0.906	0.975
	44	0.936	0.908	0.988
	32	0.936	0.906	1.000
	60	0.917	0.907	0.790
	64	0.862	0.833	0.642
	1	0.730	0.704	0.580

418 H DeepSeek positions B and C

419 I The four-disk Tower of Hanoi state space

420 The state space of the four-disk Tower of Hanoi forms a Sierpiński triangle (Figure 6): each
 421 node is a valid configuration and each edge a legal single-disk move. The three top-level
 422 sub-triangles partition the space by the position of the largest disk, the recursive cluster
 423 structure whose inflated cross-cluster separation produces the Spearman–Pearson gap
 424 reported throughout.

Table 9: Probing DeepSeek-R1-Distill-Qwen-32B through generation (layers 24/36/48); columns as in Table 5.

Position	Layer	Spearman	Acc	D0	D1	D2	D3
A	24	0.935	1.00	0.68	0.56	0.68	0.89
A	36	0.935	1.00	0.61	0.61	0.73	0.86
A	48	0.933	0.98	0.62	0.63	0.63	0.83
B	24	0.775	0.51	0.26	0.25	0.30	0.41
B	36	0.807	0.38	0.30	0.30	0.37	0.44
B	48	0.737	0.33	0.34	0.36	0.34	0.47
C	24	0.894	0.71	0.77	0.46	0.50	0.51
C	36	0.899	0.68	0.64	0.48	0.52	0.51
C	48	0.863	0.57	0.50	0.41	0.50	0.51

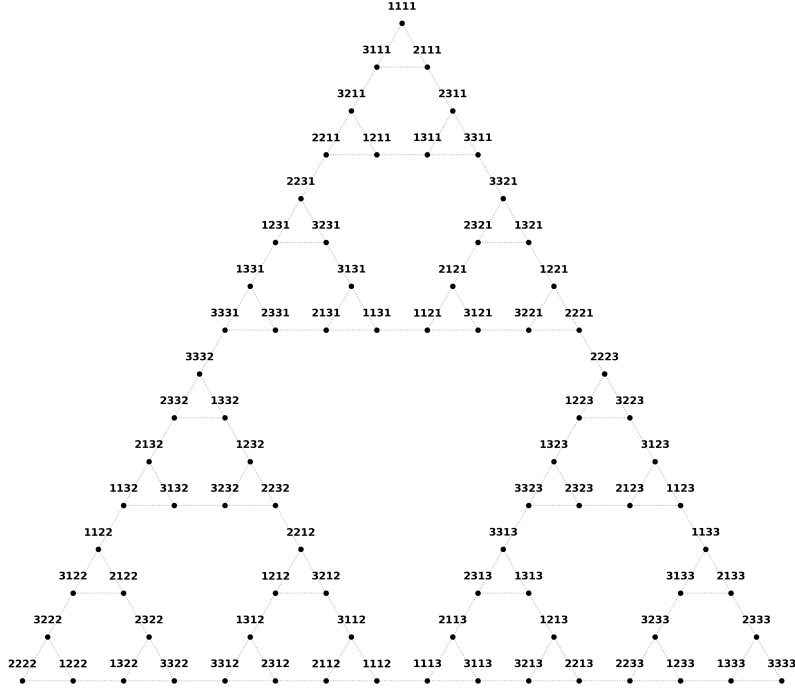


Figure 6: The state space of the 4-disk Tower of Hanoi forms a Sierpiński triangle. Each node is labelled by a triple indicating the peg on which each disk rests, ordered from smallest to largest. The three sub-triangles partition the space according to the position of the largest disk.